Mission to Mars

By Sheela Raman



It had taken decades of hard work, but at long last the day arrived. Hundreds of spectators gathered at NASA's Kennedy Air Force Base to watch the Orion 254 shuttle shoot up into outer space, headed for Mars. Almost everyone in the world had tuned in to watch live footage of the launch on their smartphones or on TV. Onboard the shuttle two astronauts, dressed in puffy white spacesuits, patiently waited for blastoff. Roy and Ciara Thomas were a married couple NASA had specially selected to undertake this risky mission. In just a few minutes, they would begin an adventure that would carry them further away from Earth than anyone else in human history.

A voice came over the speakers that hovered above the excited spectators. Suddenly the crowd hushed. "In 1969, human beings walked on the moon for the first time," said the deep, confident voice, "and now, in July of 2020, we expand our frontier to Mars. Please send your salutations and blessings to our brave astronauts as they embark on this groundbreaking mission. "

Everyone erupted into cheers and whistles, and Ciara and Roy smiled as they heard these expressions of support over their shuttle radio. Family and friends of the astronauts stood at the front of the crowd. Tears streamed down their faces. They were proud of Roy and Ciara, but they did not know when or if they would ever see them again. The voyage would take at least a year to complete, and it involved many risks. Although Earth was now at its closest possible distance to Mars, the famed Red Planet was still 34 million miles away. The moon, which is 240,000 miles away from Earth, seemed a walk around the block by comparison.

"Ten," counted the deep voice, "nine, eight, seven, six, five, four, three, two, one!" And with that, colossal fuel jets attached to the shuttle shot white-hot streaks of fire into the ground, and the shuttle and tanks immediately shot upward into the sky. In order to generate enough

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force to propel the shuttle from the earth's surface all the way into outer space, the shuttle's fuel tanks had to send an enormous amount of opposite pressure against the ground. That's why NASA scientists designed the jets to be so big and powerful. The jets created a great show of fire and smoke when they released.

Within minutes, Roy and Ciara passed through the top layers of the earth's atmosphere. The fuel jets, which were no longer necessary, broke away from the shuttle and drifted off into the vastness of outer space. Roy and Ciara began to float inside the shuttle because Earth's gravitational pull no longer weighed them down. Oxygen tanks fed air into their surroundings so that they could breathe. Roy and Ciara took their first deep breaths in space and gazed out the window. A brilliant blue sphere rose before them amid the blackness like a giant, shimmering jewel. That was Planet Earth, home to their fellow men and women, and they were now hurtling away from it at 75,000 miles per hour.

"Ground control to Roy and Ciara," said a happy voice on the radio from Mission Control.

"Roy Thomas to Mission Control," replied Roy.

"That was a textbook takeoff," the officer said. "Congratulations!"

"Thanks," said Roy. "We'll keep you posted as we move along."

NASA chose Roy and Ciara for the Mars mission not only because they were both very intelligent and physically fit, but also because they were happily married. Over such a long trip, NASA felt it was better to send people who would not easily get into fights. If anything went wrong on board the spacecraft, Roy and Ciara would have to work as a tight team to fix the problem. For an entire year they would have to survive without a single freshly cooked meal—all their food was stored on the craft in vacuum-sealed packs and tubes. A finite cargo of oxygen tanks contained their air supply. If at any point the mission stalled for too long, they ran the risk of running out of air to breathe. To maintain a good attitude amidst these challenges, the two astronauts really had to enjoy each other's company.

The risks of the Mars voyage did not seem to faze either Roy or Ciara. They were excited about the contribution they would make to human understanding of the universe. Roy was tall and broad-shouldered, and had been an Air Force captain before joining the team at NASA. Ciara was a petite, fine-boned woman, who worked as an astrophysicist for most of her career before volunteering to train as an astronaut.

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For one whole quiet year the couple sailed across space. Every now and then Roy grew claustrophobic inside the small craft, but when this happened he exercised on the special fitness machines, or donned his spacesuit and opened the door of the craft, climbing on top of its surface to check that all parts were running smoothly. Ciara wrote in a journal about the beautiful sights she saw out the window—distant galaxies and stars, a giant asteroid not so far away. By the time Orion 254 finally approached the arid, rust-colored surface of Mars, people on Earth had almost forgotten about them. But as the craft got closer and closer to the red surface, news channels on Earth started buzzing. “Humans on Mars, at long last! Another giant leap for mankind!” they proclaimed.

Roy manned the gears of the craft now, directing it to a flat plain on the surface of Mars, just between two deep craters. He was an expert pilot, and he landed the spacecraft with a gentle thud. “Bravo!” shouted Mission Control. Everyone was clapping in the background. “You’re about to make history!”

With great care and attention, the astronauts put their spacesuits back on. On top of these suits they strapped an extra layer to protect against radiation, much like a doctor puts a protective layer over a patient before performing an X-ray. Unlike Earth, Mars does not have a very thick atmosphere or an ozone layer, so the radiation from the sun is many times stronger there. In a pouch attached to her spacesuit, Ciara carried jars and a small shovel to collect samples of Martian soil. She would be the first person to bring this precious material back to Earth. What would they discover in it? Alien bacteria? Some powerful, indestructible metal?

Roy switched on the television monitors outside the spacecraft, so citizens of Earth could watch this historic, first walk on Mars. The two astronauts fixed microphones to the insides of their suits so that they could talk to each other, Mission Control, and the rest of the earthly world. They fastened their spacesuits to cords inside the craft and stepped out into the swirling dust.

As he took his first steps, Roy saw a flicker of silver out of the corner of his eye. It seemed to move through the air and settle behind a rock to his right. Forgetting for a moment that he had to move slowly, Roy dashed forward to catch a better glimpse. In that one moment the cord attaching him to the spacecraft suddenly snapped, and he went floating into the air. People watching from Earth gripped their chairs in fear. There was not enough gravity on Mars to hold Roy steadily on the ground.

Ciara saw what happened and immediately shouted to him over the intercom. “Throw something out in front of you! As hard as you can!”

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Roy knew when to listen to his wife. He removed a big hammer from his tool belt and hurled it with all his might directly ahead of him. The force of his throw generated an equal and opposite force that sent Roy hurtling backwards—right back into the spacecraft. He grabbed hold of the doors of the craft to steady himself and then looked at the broken cord. Fortunately they had the materials to fix the tear, and Ciara and Roy worked together quickly to patch it up. Within fifteen minutes they were back out on the surface. Mission Control and most of planet Earth cheered them along.

“Mars landing, take two,” said Roy, to Mission Control.

“What happened?” asked the commander at Mission Control. “Why did you rush forward like that?”

“You’re not going to believe this,” said Roy. “But I swear I saw something silver, moving behind that rock just ahead.”

“Are you saying . . . that you might have seen a life form?” asked the commander in a serious, quiet voice.

“Could be,” said Roy. “Won’t know for sure until we explore a bit.”

“Indeed,” said Ciara. “We’ll have to be very quiet and slow.”

Side by side, tools in hand, Roy and Ciara Thomas ventured forth into the mysterious red landscape.

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Name: _____ **Date:** _____

1. Where is the shuttle carrying Roy and Ciara going?

A) the moon
B) the sun
C) Mars
D) Venus

2. What is the climax of the action in this story?

A) The cord connecting Roy to the spacecraft snaps.
B) People from around the world watch as the Orion 254 shuttle takes off.
C) The fuel jets on the spacecraft drift off into outer space.
D) Roy gets claustrophobic while on the spacecraft.

3. Roy and Ciara work well as a team.
What evidence from the passage supports this statement?

A) Ciara is a petite, fine-boned woman who worked as an astrophysicist before becoming an astronaut.
B) Sometimes Roy exercises on special fitness machines when he gets claustrophobic on the spacecraft.
C) After his cord snaps, Roy saves himself by following the instructions that Ciara gives him.
D) Ciara says that she and Roy will have to be quiet and slow when they explore Mars.

4. How do people on Earth feel about the mission to Mars?

A) People on Earth only become interested in the mission during the landing on Mars.
B) People on Earth are interested in the mission during takeoff and landing, but they lose interest in between.
C) People on Earth are not interested in the mission during takeoff and landing, but they become interested in between.
D) People on Earth are interested in the mission during takeoff, landing, and every point in between.

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5. What is a theme of this story?

A) the pleasures of old age
B) the importance of recycling
C) the challenges of living in a big city
D) the excitement of discovery

6. Read the following sentences: "'That was a **textbook takeoff**,' the officer said. 'Congratulations!'" 'Thanks,' said Roy. 'We'll keep you posted as we move along.'" What does the phrase **textbook takeoff** mean above?

A) a fuel jet that did not work they way it was supposed to
B) a launch that went exactly the way it was supposed to
C) an astronaut who likes to write about stars and asteroids
D) a group of people who gather together for an important event

7. Choose the answer that best completes the sentence below.
Roy dashes forward _____ he sees something silver move through the air on Mars.

A) after
B) before
C) as an illustration
D) thus

8. Who are Roy and Ciara Thomas?

9. Why did NASA choose Roy and Ciara for the Mars mission?

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10. Were Roy and Ciara a good choice for the Mars mission? Explain why or why not, using evidence from the story.

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Teacher Guide & Answers

Passage Reading Level: Lexile 1030

1. **C** *Mars*

2. **A** *The cord connecting Roy to the spacecraft snaps.*

3. **C** *After his cord snaps, Roy saves himself by following the instructions that Ciara gives him.*

4. **B** *People on Earth are interested in the mission during takeoff and landing, but they lose interest in between.*

5. **D** *the excitement of discovery*

6. **B** *a launch that went exactly the way it was supposed to*

7. **A** *after*

8. **Suggested answer:** *(answers may vary but should resemble the following): Roy and Ciara Thomas are a married couple chosen by NASA for a mission to Mars.*

9. **Suggested answer:** *NASA chose Roy and Ciara for the Mars mission because they are intelligent, physically fit, and happily married.*

10. **Suggested answer:** *Answers may vary, as long as they are supported by the story. Students may argue that Roy and Ciara were a good choice for the Mars mission, citing such evidence as their ability to work together during the emergency of Roy's cord snapping. On the other hand, students may argue that Roy was not a good choice for the Mars mission, as he impulsively dashes forward after seeing something silver on Mars, causing the cord connecting him with the spacecraft to snap.*

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